

Module 6 — Claude Agent SDK

Lesson 6.1

Claude Agent SDK overview

The SDK is Claude Code's engine, exposed as a library. Use it when you want a Claude-Code-style agent without the interactive UI.

Why it matters

Claude Code is great for interactive work. But many useful workflows aren't interactive: scheduled triage of new issues, post-commit changelog generation, an inbox monitor that drafts replies, a CI job that auto-fixes lint failures. The Claude Agent SDK is how you build those.

The same model, same tools, same plan/explore/implement patterns — packaged as code you run yourself.

SDK vs. CLI vs. API

Three ways to drive Claude. Pick by use case:

Tool	Use when
Claude Code (CLI)	Interactive coding sessions, IDE work
Claude Agent SDK	Headless agents — scripts, cron jobs, CI, services
Raw Anthropic API	You don't need agent behavior; just one model call

The SDK is the right layer when you want:

- The agentic loop (model decides on tool calls, system loops until done)
- Built-in file/edit/bash/search tools without re-implementing them
- MCP server support
- A managed conversation lifecycle

If you only want "generate text from prompt," use the raw API directly.

Available packages

Language	Package
Python	<code>claude-agent-sdk</code>
TypeScript	<code>@anthropic-ai/claude-agent-sdk</code>

This module focuses on Python; TypeScript usage is structurally similar.

What the SDK gives you

- A high-level `query()` / agent runner that handles the model + tool loop
- Built-in tools matching what Claude Code uses (Read, Write, Edit, Glob, Grep, Bash)
- MCP server integration (connect your agent to MCP just like Claude Code does)
- Streaming responses, tool-call events, usage metrics
- Optional permission callbacks for tool gating

When SDK is the wrong tool

- **One-off prompt** — just use the Anthropic SDK directly

- **A web chat** — Claude.ai or your own UI on the API
- **Multi-agent orchestration with deep state** — you may need more control than the SDK exposes (build on the raw API)

What the example will do

examples/05-sdk-agent-python is a small headless agent:

- Takes a prompt from the CLI
- Runs against a sandbox directory
- Has Read/Edit/Bash tools available
- Prints structured output (tool calls, model text, final result)

It's the smallest useful agent — about 50 lines. From there you can grow it into a cron job, a webhook handler, a CI step, whatever.

The mental model

```
your script ■■■ ClaudeAgentSDK ■■■ Claude API
                ■
                ■■■■ Built-in tools (Read/Edit/Bash/Glob/Grep)
                ■■■■ MCP servers (optional)
                ■■■■ Your custom tools (optional)
```

You decide:

- System prompt
- Initial user message (the task)
- Allowed tools and permissions
- Working directory
- Whether to stream or get-all-at-end

The SDK runs the model-tool loop until the task is done (or you cap it).

Prereqs

- An Anthropic API key (`ANTHROPIC_API_KEY`)
- Python 3.11+ (for the Python SDK)

Pricing reality check

Agent runs are not single API calls — they're loops with multiple turns and tool calls. A simple agent task might be 10–50 model turns. Budget accordingly:

- For dev/test: a few cents per run is normal
- For production: use prompt caching (covered in 6.4) to cut costs dramatically
- For scale: budget per agent run × invocations/day; consider Haiku for narrow tasks

Try it

Skim ahead to 6.2 Your first headless agent and the runnable examples/05-sdk-agent-python. Even just reading the code clarifies the model.

Gotchas

- **SDK and Code share concepts but not config files.** A `.claude/settings.json` in your project doesn't automatically apply to SDK runs — you configure permissions through SDK options.
- **API key auth** — the SDK doesn't use Claude.ai subscription auth (that's Code-only). You need an API key.
- **Costs grow with autonomy.** A loop with a lot of tool calls can be expensive. Always cap max turns in production.

Further reading

- Python SDK: <https://github.com/anthropics/claude-agent-sdk-python>
- TypeScript SDK: <https://github.com/anthropics/claude-agent-sdk-typescript>
- Official Agent SDK docs: <https://docs.claude.com/en/docs/agents/agent-sdk>

Next

→ 6.2 Your first headless agent